

EXPERIMENT NO. 4

**DESIGN AND IMPLEMENTATION OF EDGE–CLOUD
SYNCHRONIZATION USING LOCAL MQTT BRIDGE (GOOGLE
COLAB VERSION)**

Aim

To simulate Edge–Cloud synchronization by collecting IoT sensor data at an edge device, synchronizing it to the cloud, and performing real-time analytics using Python in Google Colab.

Objectives

- Simulate an Edge Device.
- Generate real-time IoT sensor data.
- Synchronize data from Edge to Cloud.
- Store synchronized cloud data.
- Perform stream analytics.
- Detect abnormal conditions.
- Visualize synchronized data.

Software required

- Google Colab
- Python 3
- pandas
- numpy
- matplotlib

Theory

- Edge Computing processes data close to the IoT device, reducing latency.
- Cloud Computing stores and analyzes synchronized data for long-term monitoring.

Data Flow:

IoT Device



Edge Device



MQTT Bridge (Simulation)



Cloud Storage



Dashboard

Scenario

- Smart Water Tank Monitoring
- Edge Device collects water level.
- Cloud stores synchronized values.
- Alert generated if water level is very low.

Algorithm

- Start Program.
- Generate Water Tank Level.
- Store data in Edge Memory.
- Synchronize data to Cloud.
- Calculate Mean.
- Calculate Moving Average.
- Calculate Standard Deviation.
- Detect Low Water Level.
- Display Graph.
- Save CSV.

Google colab code

```
import random

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

from IPython.display import clear_output

import time

edge_storage = []

cloud_storage = []

time_data = []

for i in range(1,51):

    water_level = random.randint(10,100)

    # Edge Storage

    edge_storage.append(water_level)

    # Synchronize to Cloud

    cloud_storage.append(water_level)
```

```

time_data.append(i)
mean = np.mean(cloud_storage)
std = np.std(cloud_storage)
if len(cloud_storage)>=5:
    moving=np.mean(cloud_storage[-5:])
else:
    moving=mean
if water_level<20:
    status="LOW WATER LEVEL"
else:
    status="NORMAL"
clear_output(wait=True)
print("===== EDGE DEVICE =====")
print("Current Water Level :",water_level,"%")
print("Stored at Edge :",len(edge_storage))
print()
print("===== CLOUD =====")
print("Cloud Records :",len(cloud_storage))
print("Mean :",round(mean,2))

print("Moving Average :",round(moving,2))
print("Standard Deviation :",round(std,2))
print("Status :",status)
plt.figure(figsize=(10,4))
plt.plot(time_data,cloud_storage,marker='o')
plt.axhline(20,color='red',linestyle='--',label='Critical Level')
plt.xlabel("Sample")
plt.ylabel("Water Level (%)")
plt.title("Edge-Cloud Synchronization")
plt.legend()

```

```

plt.grid(True)

plt.show()

time.sleep(0.5)

df=pd.DataFrame({

    "Sample":time_data,

    "Water Level":cloud_storage

})

df.to_csv("cloud_water_data.csv",index=False)

print("Experiment Completed Successfully")

```

OUTPUT

===== EDGE DEVICE =====

Current Water Level : 18 %

Stored at Edge : 32

===== CLOUD =====

Cloud Records : 32

Mean : 54.81

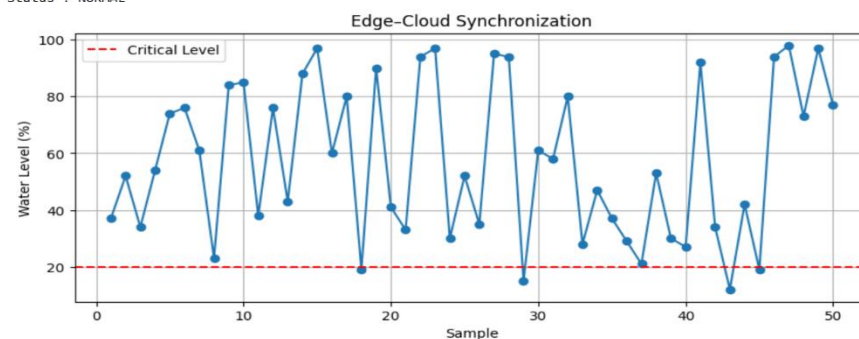
Moving Average : 29.20

Standard Deviation : 21.65

Status : LOW WATER
LEVEL

===== EDGE DEVICE =====
Current Water Level : 77 %
Stored at Edge : 50

===== CLOUD =====
Cloud Records : 50
Mean : 57.32
Moving Average : 87.8
Standard Deviation : 27.14
Status : NORMAL



Experiment Completed Successfully

Result

The edge device successfully collected water level data and synchronized it with the cloud. The cloud performed stream analytics using Mean, Moving Average, and Standard Deviation, detected low water level conditions, and stored the synchronized data for future analysis.